Web Programming Lab 3 - Recitation

Question 1

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>AI Art Studio - Pricing Plans</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }
        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            overflow-x: hidden;
        }
        .split-container {
            display: flex;
            min-height: 100vh;
        }
        .hero-section {
            flex: 1;
            background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
            padding: 60px;
            color: white;
            display: flex;
            flex-direction: column;
            justify-content: center;
        }
        .hero-title {
            font-size: 4em;
            margin-bottom: 15px;
            text-shadow: 2px 2px 4px rgba(0,0,0,0.2);
        }
        .hero-subtitle {
            font-size: 1.5em;
            opacity: 0.95;
            margin-bottom: 40px;
        }
        .art-grid {
            display: grid;
            grid-template-columns: repeat(2, 1fr);
            gap: 20px;
            margin-bottom: 30px;
        }
        .art-item {
            aspect-ratio: 1;
            border-radius: 15px;
            overflow: hidden;
            box-shadow: 0 10px 30px rgba(0,0,0,0.3);

            transition: transform 0.3s ease;
        }
        .art-item:hover {
            transform: scale(1.05);
        }
        .art-item img {
            width: 100%;
            height: 100%;
            object-fit: cover;
        }
        .art-features {
            display: flex;
            gap: 15px;
            flex-wrap: wrap;
        }
        .feature-badge {
            background: rgba(255,255,255,0.2);
            backdrop-filter: blur(10px);
            padding: 12px 20px;
```

```
    border-radius: 25px;
    font-size: 1.05em;
}
.pricing-section {
    flex: 1;
    padding: 60px;
    background: #f8f9fa;
    overflow-y: auto;
}
.pricing-section h1 {
    font-size: 3em;
    color: #2c3e50;
    margin-bottom: 40px;
    text-align: center;
}
.pricing-container {
    display: flex;
    flex-direction: column;
    gap: 25px;
}
.pricing-card {
    background: white;
    border-radius: 20px;
    padding: 35px;
    box-shadow: 0 5px 20px rgba(0,0,0,0.1);
    transition: transform 0.3s ease, box-shadow 0.3s ease;
    border-left: 5px solid #667eea;
}
.pricing-card:hover {
    transform: translateX(10px);
    box-shadow: 0 10px 30px rgba(0,0,0,0.15);

}
.pricing-card.pro-plan {
    border-left-color: #f093fb;
    position: relative;
}
.pricing-card.pro-plan:before {
    content: '■ POPULAR';
    position: absolute;
    top: -15px;
    right: 30px;
    background: linear-gradient(135deg, #f093fb 0%, #f5576c 100%);
    color: white;
    padding: 5px 15px;
    border-radius: 15px;
    font-size: 0.85em;
    font-weight: bold;
}
.pricing-card.enterprise-plan {
    border-left-color: #4facfe;
}
.pricing-card h2 {
    font-size: 2em;
    color: #2c3e50;
    margin-bottom: 10px;
}
.price {
    font-size: 2.5em;
    color: #667eea;
    font-weight: bold;
    margin-bottom: 20px;
}
.pricing-card.pro-plan .price {
    color: #f093fb;
}
.pricing-card.enterprise-plan .price {
    color: #4facfe;
}
.features {
    list-style: none;
    margin-bottom: 25px;
}
.features li {
    padding: 12px 0;
    border-bottom: 1px solid #ecf0f1;
    display: flex;
    align-items: center;
    gap: 10px;
}
.features li:before {
    content: '✓';

    color: #27ae60;
    font-weight: bold;
```

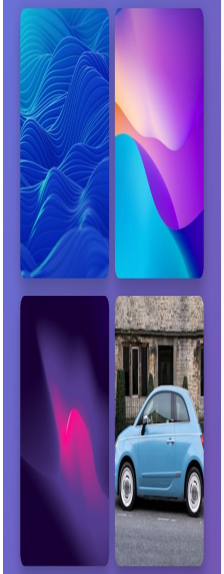
```
            font-size: 1.2em;
        }
        .btn {
            width: 100%;
            padding: 15px;
            background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
            color: white;
            border: none;
            border-radius: 10px;
            font-size: 1.1em;
            font-weight: bold;
            cursor: pointer;
            transition: all 0.3s ease;
        }
        .btn:hover {
            transform: translateY(-3px);
            box-shadow: 0 10px 20px rgba(102, 126, 234, 0.3);
        }
        .pricing-card.pro-plan .btn {
            background: linear-gradient(135deg, #f093fb 0%, #f5576c 100%);
        }
        .pricing-card.enterprise-plan .btn {
            background: linear-gradient(135deg, #4facfe 0%, #00f2fe 100%);
        }
        @media (max-width: 1024px) {
            .split-container {
                flex-direction: column;
            }
        }
    </style>
</head>
<body>
    <div class="split-container">
        <!-- Left Side: Hero/Art Section -->
        <div class="hero-section">
            <div class="hero-content">
                <h1 class="hero-title">■ AI Art Studio</h1>
                <p class="hero-subtitle">Create stunning artwork with the power of AI</p>
                <div class="art-showcase">
                    <div class="art-grid">
                        <div class="art-item">
                            <img src="https://images.unsplash.com/photo-1634017839464-5c339ebe3cb4?w=400&q=80" al
                        </div>
                        <div class="art-item">
                            <img src="https://images.unsplash.com/photo-1618005182384-a83a8bd57fbe?w=400&q=80" al
                        </div>
                        <div class="art-item">
                            <img src="https://images.unsplash.com/photo-1620641788421-7a1c342ea42e?w=400&q=80" al

                        </div>
                        <div class="art-item">
                            <img src="https://images.unsplash.com/photo-1549317661-bd32c8ce0db2?w=400&q=80" alt="A
                        </div>
                    </div>
                    <div class="art-features">
                        <div class="feature-badge">■ Unlimited Generations</div>
                        <div class="feature-badge">■ AI-Powered Creation</div>
                        <div class="feature-badge">■■ High-Resolution Output</div>
                    </div>
                </div>
            </div>
        </div>

        <!-- Right Side: Pricing Section -->
        <div class="pricing-section">
            <h1>■ Pricing Plans</h1>

            <div class="pricing-container">
                <div class="pricing-card basic-plan">
                    <h2>Basic</h2>
                    <p class="price">$9<span style="font-size:0.5em">/month</span></p>
                    <ul class="features">
                        <li>100 AI Generations/month</li>
                        <li>Standard Resolution (1024x1024)</li>
                        <li>Basic Styles & Filters</li>
                        <li>Email Support</li>
                        <li>Personal Use Only</li>
                    </ul>
                    <button class="btn">Get Started</button>
                </div>

                <div class="pricing-card pro-plan">
                    <h2>Professional</h2>
                    <p class="price">$29<span style="font-size:0.5em">/month</span></p>
                    <ul class="features">
```

```
                        <li>1000 AI Generations/month</li>
                        <li>High Resolution (2048x2048)</li>
                        <li>All Styles & Advanced Filters</li>
                        <li>Priority Support</li>
                        <li>Commercial Use License</li>
                        <li>Custom Training Models</li>
                    </ul>
                    <button class="btn">Choose Pro</button>
                </div>

                <div class="pricing-card enterprise-plan">
                    <h2>Enterprise</h2>
                    <p class="price">$99<span style="font-size:0.5em">/month</span></p>
                    <ul class="features">

                        <li>Unlimited AI Generations</li>
                        <li>Ultra HD Resolution (4096x4096)</li>
                        <li>All Features + Custom Styles</li>
                        <li>24/7 Dedicated Support</li>
                        <li>Full Commercial License</li>
                        <li>Private Training Models</li>
                        <li>API Access</li>
                        <li>Team Collaboration Tools</li>
                    </ul>
                    <button class="btn">Contact Sales</button>
                </div>
            </div>
        </div>
    </div>
</body>
</html>
```

Output:

🎨 AI Art Studio

Create stunning artwork with the power of AI



⚡ Unlimited Generations

🎨 AI-Powered Creation

🖼️ High-Resolution Output

💎 Pricing Plans

Basic

$9/month

- ✓ 100 AI Generations/month
- ✓ Standart Resolution (1024x1024)
- ✓ Basic Styles & Filters
- ✓ Email Support
- ✓ Personal Use Only

Get Started

★ POPULAR

Professional

$29/month

- ✓ 1000 AI Generations/month
- ✓ High Resolution (2048x2048)
- ✓ All Styles & Advanced Filters
- ✓ Priority Support
- ✓ Commercial Use License
- ✓ Custom Training Models

Choose Pro

Enterprise

$99/month

- ✓ Unlimited AI Generations
- ✓ Ultra HD Resolution (4096x4096)
- ✓ All Features + Custom Styles
- ✓ 24/7 Dedicated Support
- ✓ Full Commercial License
- ✓ Private Training Models
- ✓ API Access
- ✓ Team Collaboration Tools

Contact Sales

Question 2

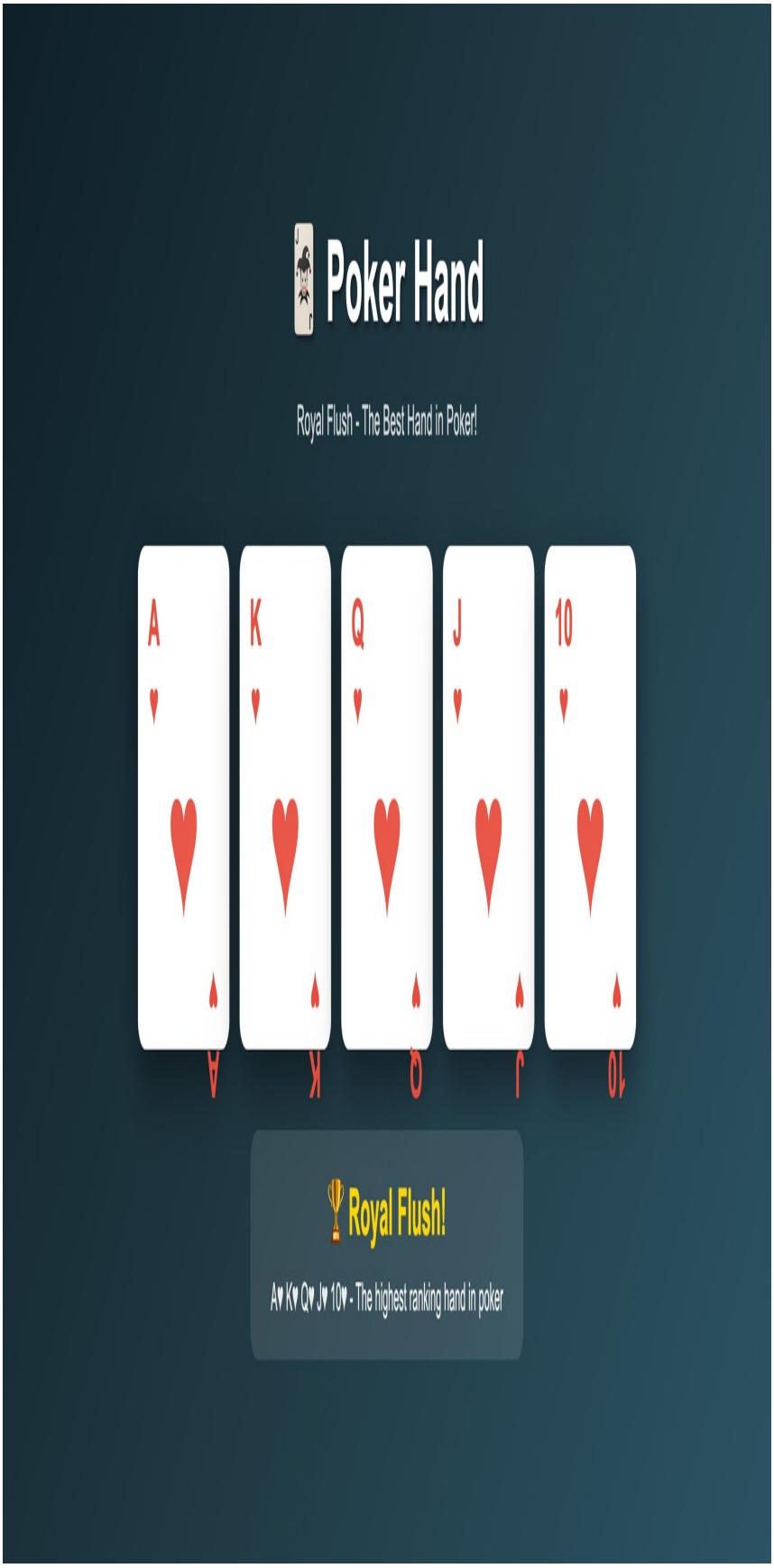
Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Poker Hand - Playing Cards</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }
        body {
            background: linear-gradient(135deg, #0F2027 0%, #203A43 50%, #2C5364 100%);
            font-family: 'Arial', sans-serif;
            padding: 50px 20px;
            min-height: 100vh;
            display: flex;
            flex-direction: column;
            align-items: center;
            justify-content: center;
        }
        h1 {
            color: white;
            font-size: 3.5em;
            text-align: center;
            margin-bottom: 20px;
            text-shadow: 2px 2px 4px rgba(0,0,0,0.5);
        }
        .subtitle {
            color: rgba(255,255,255,0.8);
            font-size: 1.3em;
            text-align: center;
            margin-bottom: 50px;
        }
        #playing-cards {
            display: flex;
            gap: 20px;
            flex-wrap: wrap;
            justify-content: center;
            perspective: 1000px;
        }
        .card {
            width: 180px;
            height: 250px;
            background: white;
            border-radius: 15px;
            padding: 20px;
            display: flex;
            flex-direction: column;

            justify-content: space-between;
            box-shadow: 0 15px 40px rgba(0,0,0,0.5);
            transition: all 0.3s ease;
            cursor: pointer;
            position: relative;
        }
        .card:hover {
            transform: translateY(-20px) rotateY(10deg);
            box-shadow: 0 25px 60px rgba(0,0,0,0.6);
        }
        .card:before {
            content: '';
            position: absolute;
            top: 0;
            left: 0;
            right: 0;
            bottom: 0;
            background: linear-gradient(135deg, transparent 0%, rgba(255,255,255,0.1) 100%);
            border-radius: 15px;
            pointer-events: none;
        }
        .card .left,
        .card .right {
            display: flex;
            flex-direction: column;
            align-items: center;
            gap: 5px;
```

```
        }
        .card .left {
            align-self: flex-start;
        }
        .card .right {
            align-self: flex-end;
            transform: rotate(180deg);
        }
        .card .middle {
            flex: 1;
            display: flex;
            align-items: center;
            justify-content: center;
        }
        .rank {
            font-size: 2em;
            font-weight: bold;
        }
        .suit {
            font-size: 1.8em;
        }
        .suit-large {
            font-size: 6em;

        }
        .hearts .rank,
        .hearts .suit,
        .hearts .suit-large {
            color: #e74c3c;
        }
        .diamonds .rank,
        .diamonds .suit,
        .diamonds .suit-large {
            color: #e74c3c;
        }
        .spades .rank,
        .spades .suit,
        .spades .suit-large {
            color: #2c3e50;
        }
        .clubs .rank,
        .clubs .suit,
        .clubs .suit-large {
            color: #2c3e50;
        }
        .hand-info {
            text-align: center;
            margin-top: 40px;
            background: rgba(255,255,255,0.1);
            backdrop-filter: blur(10px);
            padding: 20px 40px;
            border-radius: 15px;
            color: white;
        }
        .hand-info h2 {
            font-size: 2em;
            margin-bottom: 10px;
            color: #ffd700;
        }
        .hand-info p {
            font-size: 1.2em;
            opacity: 0.9;
        }
        @media (max-width: 768px) {
            #playing-cards {
                gap: 15px;
            }
            .card {
                width: 140px;
                height: 200px;
                padding: 15px;
            }
            .suit-large {
                font-size: 4em !important;

            }
        }
    </style>
</head>
<body>
    <h1>■ Poker Hand</h1>
    <p class="subtitle">Royal Flush - The Best Hand in Poker!</p>

    <div id="playing-cards">
        <!-- Ace of Hearts -->
```

```
        <div class="card hearts">
            <div class="left">
                <div class="rank">A</div>
                <div class="suit">♥</div>
            </div>
            <div class="middle">
                <div class="suit-large">♥</div>
            </div>
            <div class="right">
                <div class="rank">A</div>
                <div class="suit">♥</div>
            </div>
        </div>

        <!-- King of Hearts -->
        <div class="card hearts">
            <div class="left">
                <div class="rank">K</div>
                <div class="suit">♥</div>
            </div>
            <div class="middle">
                <div class="suit-large">♥</div>
            </div>
            <div class="right">
                <div class="rank">K</div>
                <div class="suit">♥</div>
            </div>
        </div>

        <!-- Queen of Hearts -->
        <div class="card hearts">
            <div class="left">
                <div class="rank">Q</div>
                <div class="suit">♥</div>
            </div>
            <div class="middle">
                <div class="suit-large">♥</div>
            </div>
            <div class="right">
                <div class="rank">Q</div>

                <div class="suit">♥</div>
            </div>
        </div>

        <!-- Jack of Hearts -->
        <div class="card hearts">
            <div class="left">
                <div class="rank">J</div>
                <div class="suit">♥</div>
            </div>
            <div class="middle">
                <div class="suit-large">♥</div>
            </div>
            <div class="right">
                <div class="rank">J</div>
                <div class="suit">♥</div>
            </div>
        </div>

        <!-- 10 of Hearts -->
        <div class="card hearts">
            <div class="left">
                <div class="rank">10</div>
                <div class="suit">♥</div>
            </div>
            <div class="middle">
                <div class="suit-large">♥</div>
            </div>
            <div class="right">
                <div class="rank">10</div>
                <div class="suit">♥</div>
            </div>
        </div>
    </div>

    <div class="hand-info">
        <h2>■ Royal Flush!</h2>
        <p>A♥ K♥ Q♥ J♥ 10♥ - The highest ranking hand in poker</p>
    </div>
</body>
</html>
```

Output:


Poker Hand
Royal Flush - The Best Hand in Poker!
A
K
Q
J
10
Royal Flush!
A♥ K♥ Q♥ J♥ 10♥ - The highest ranking hand in poker

Question 3

Code:

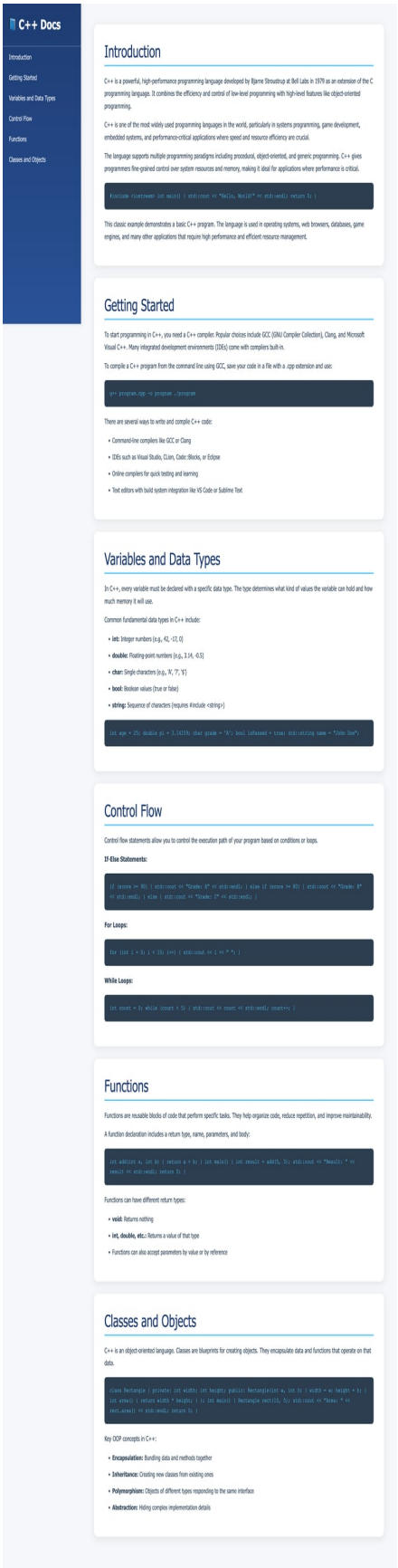
```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>C++ Documentation</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }
        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            background: #f5f7fa;
            display: flex;
            min-height: 100vh;
        }
        #navbar {
            width: 300px;
            background: linear-gradient(180deg, #1e3c72 0%, #2a5298 100%);
            color: white;
            position: fixed;
            height: 100vh;
            overflow-y: auto;
            box-shadow: 3px 0 10px rgba(0,0,0,0.1);
        }
        #navbar header {
            font-size: 2em;
            padding: 30px 20px;
            border-bottom: 2px solid rgba(255,255,255,0.2);
            font-weight: bold;
        }
        .nav-link {
            display: block;
            color: rgba(255,255,255,0.9);
            text-decoration: none;
            padding: 15px 20px;
            border-left: 4px solid transparent;
            transition: all 0.3s ease;
        }
        .nav-link:hover {
            background: rgba(255,255,255,0.1);
            border-left-color: #4fc3f7;
            padding-left: 25px;
        }
        #main-doc {
            margin-left: 300px;
            padding: 50px;
            flex: 1;

        }
        .main-section {
            background: white;
            border-radius: 15px;
            padding: 40px;
            margin-bottom: 30px;
            box-shadow: 0 5px 20px rgba(0,0,0,0.08);
        }
        .main-section header {
            font-size: 2.5em;
            color: #1e3c72;
            margin-bottom: 20px;
            border-bottom: 3px solid #4fc3f7;
            padding-bottom: 10px;
        }
        .main-section p {
            line-height: 1.8;
            color: #34495e;
            margin-bottom: 15px;
            font-size: 1.05em;
        }
        code {
            display: block;
            background: #2c3e50;
            color: #4fc3f7;
            padding: 20px;
            border-radius: 8px;
```

```
            margin: 20px 0;
            font-family: 'Courier New', monospace;
            overflow-x: auto;
            font-size: 0.95em;
            line-height: 1.6;
        }
        ul, ol {
            margin: 15px 0 15px 30px;
            line-height: 2;
            color: #34495e;
        }
        ul li, ol li {
            margin: 8px 0;
        }
        @media (max-width: 768px) {
            #navbar {
                width: 100%;
                position: relative;
                height: auto;
            }
            #main-doc {
                margin-left: 0;
                padding: 20px;

            }
        }
    </style>
</head>
<body>
    <!-- Navigation Bar -->
    <nav id="navbar">
        <header>■ C++ Docs</header>
        <a href="#Introduction" class="nav-link">Introduction</a>
        <a href="#Getting_Started" class="nav-link">Getting Started</a>
        <a href="#Variables_and_Data_Types" class="nav-link">Variables and Data Types</a>
        <a href="#Control_Flow" class="nav-link">Control Flow</a>
        <a href="#Functions" class="nav-link">Functions</a>
        <a href="#Classes_and_Objects" class="nav-link">Classes and Objects</a>
    </nav>

    <!-- Main Documentation -->
    <main id="main-doc">
        <!-- Section 1: Introduction -->
        <section class="main-section" id="Introduction">
            <header>Introduction</header>
            <p>C++ is a powerful, high-performance programming language developed by Bjarne Stroustrup at Bell Lab
            <p>C++ is one of the most widely used programming languages in the world, particularly in systems prog
            <p>The language supports multiple programming paradigms including procedural, object-oriented, and gen
            <code>#include &lt;iostream&gt;

int main() {
    std::cout &lt;&lt; "Hello, World!" &lt;&lt; std::endl;
    return 0;
}</code>
            <p>This classic example demonstrates a basic C++ program. The language is used in operating systems, w
        </section>

        <!-- Section 2: Getting Started -->
        <section class="main-section" id="Getting_Started">
            <header>Getting Started</header>
            <p>To start programming in C++, you need a C++ compiler. Popular choices include GCC (GNU Compiler Co
            <p>To compile a C++ program from the command line using GCC, save your code in a file with a .cpp exte
            <code>g++ program.cpp -o program
./program</code>
            <p>There are several ways to write and compile C++ code:</p>
            <ul>
                <li>Command-line compilers like GCC or Clang</li>
                <li>IDEs such as Visual Studio, CLion, Code::Blocks, or Eclipse</li>
                <li>Online compilers for quick testing and learning</li>
                <li>Text editors with build system integration like VS Code or Sublime Text</li>
            </ul>
        </section>

        <!-- Section 3: Variables and Data Types -->

        <section class="main-section" id="Variables_and_Data_Types">
            <header>Variables and Data Types</header>
            <p>In C++, every variable must be declared with a specific data type. The type determines what kind of
            <p>Common fundamental data types in C++ include:</p>
            <ul>
                <li><strong>int:</strong> Integer numbers (e.g., 42, -17, 0)</li>
                <li><strong>double:</strong> Floating-point numbers (e.g., 3.14, -0.5)</li>
                <li><strong>char:</strong> Single characters (e.g., 'A', '7', '$')</li>
                <li><strong>bool:</strong> Boolean values (true or false)</li>
                <li><strong>string:</strong> Sequence of characters (requires #include &lt;string&gt;)</li>
```

```
            </ul>
            <code>int age = 25;
double pi = 3.14159;
char grade = 'A';
bool isPassed = true;
std::string name = "John Doe";</code>
        </section>

        <!-- Section 4: Control Flow -->
        <section class="main-section" id="Control_Flow">
            <header>Control Flow</header>
            <p>Control flow statements allow you to control the execution path of your program based on condition
            <p><strong>If-Else Statements:</strong></p>
            <code>if (score >= 90) {
    std::cout &lt;&lt; "Grade: A" &lt;&lt; std::endl;
} else if (score >= 80) {
    std::cout &lt;&lt; "Grade: B" &lt;&lt; std::endl;
} else {
    std::cout &lt;&lt; "Grade: C" &lt;&lt; std::endl;
}</code>
            <p><strong>For Loops:</strong></p>
            <code>for (int i = 0; i &lt; 10; i++) {
    std::cout &lt;&lt; i &lt;&lt; " ";
}</code>
            <p><strong>While Loops:</strong></p>
            <code>int count = 0;
while (count &lt; 5) {
    std::cout &lt;&lt; count &lt;&lt; std::endl;
    count++;
}</code>
        </section>

        <!-- Section 5: Functions -->
        <section class="main-section" id="Functions">
            <header>Functions</header>
            <p>Functions are reusable blocks of code that perform specific tasks. They help organize code, reduce
            <p>A function declaration includes a return type, name, parameters, and body:</p>
            <code>int add(int a, int b) {
    return a + b;
}

int main() {
    int result = add(5, 3);
    std::cout &lt;&lt; "Result: " &lt;&lt; result &lt;&lt; std::endl;
    return 0;
}</code>
            <p>Functions can have different return types:</p>
            <ul>
                <li><strong>void:</strong> Returns nothing</li>
                <li><strong>int, double, etc.:</strong> Returns a value of that type</li>
                <li>Functions can also accept parameters by value or by reference</li>
            </ul>
        </section>

        <!-- Section 6: Classes and Objects -->
        <section class="main-section" id="Classes_and_Objects">
            <header>Classes and Objects</header>
            <p>C++ is an object-oriented language. Classes are blueprints for creating objects. They encapsulate
            <code>class Rectangle {
private:
    int width;
    int height;

public:
    Rectangle(int w, int h) {
        width = w;
        height = h;
    }

    int area() {
        return width * height;
    }
};

int main() {
    Rectangle rect(10, 5);
    std::cout &lt;&lt; "Area: " &lt;&lt; rect.area() &lt;&lt; std::endl;
    return 0;
}</code>
            <p>Key OOP concepts in C++:</p>
            <ul>
                <li><strong>Encapsulation:</strong> Bundling data and methods together</li>
                <li><strong>Inheritance:</strong> Creating new classes from existing ones</li>
                <li><strong>Polymorphism:</strong> Objects of different types responding to the same interface</li
                <li><strong>Abstraction:</strong> Hiding complex implementation details</li>
```

```
            </ul>
        </section>
    </main>
</body>
</html>
```

Output:



C++ Docs

Introduction
Getting Started
Variables and Data Types
Control Flow
Functions
Classes and Objects

Introduction

C++ is a powerful, high-performance programming language developed by Bjarne Stroustrup at Bell Labs in 1979 as an extension of the C programming language. It combines the efficiency and control of low-level programming with high-level features like object-oriented programming.

C++ is one of the most widely used programming languages in the world, particularly in systems programming, game development, embedded systems, and performance-critical applications where speed and resource efficiency are crucial.

The language supports multiple programming paradigms including procedural, object-oriented, and generic programming. C++ gives programmers fine-grained control over system resources and memory, making it ideal for applications where performance is critical.

```
#include <iostream> int main() { std::cout << "Hello, World!" << std::endl; return 0; }
```

This classic example demonstrates a basic C++ program. The language is used in operating systems, web browsers, databases, game engines, and many other applications that require high performance and efficient resource management.

Getting Started

To start programming in C++, you need a C++ compiler. Popular choices include GCC (GNU Compiler Collection), Clang, and Microsoft Visual C++. Many integrated development environments (IDEs) come with compilers built-in.

To compile a C++ program from the command line using GCC, save your code in a file with a .cpp extension and use:

```
g++ program.cpp -o program ./program
```

There are several ways to write and compile C++ code:

- Command-line compilers like GCC or Clang
- IDEs such as Visual Studio, CLion, Code::Blocks, or Eclipse
- Online compilers for quick testing and learning
- Text editors with build system integration like VS Code or Sublime Text

Variables and Data Types

In C++, every variable must be declared with a specific data type. The type determines what kind of values the variable can hold and how much memory it will use.

Common fundamental data types in C++ include:

- **int:** Integer numbers (e.g., 42, -17, 0)
- **double:** Floating-point numbers (e.g., 3.14, -0.5)
- **char:** Single characters (e.g., 'A', '7', '$')
- **bool:** Boolean values (true or false)
- **string:** Sequence of characters (requires #include <string>)

```
int age = 25; double pi = 3.14159; char grade = 'A'; bool isPassed = true; std::string name = "John Doe";
```

Control Flow

Control flow statements allow you to control the execution path of your program based on conditions or loops.

If-Else Statements:

```
if (score >= 90) { std::cout << "Grade: A" << std::endl; } else if (score >= 80) { std::cout << "Grade: B"
<< std::endl; } else { std::cout << "Grade: C" << std::endl; }
```

For Loops:

```
for (int i = 0; i < 10; i++) { std::cout << i << " "; }
```

While Loops:

```
int count = 0; while (count < 5) { std::cout << count << std::endl; count++; }
```

Functions

Functions are reusable blocks of code that perform specific tasks. They help organize code, reduce repetition, and improve maintainability.

A function declaration includes a return type, name, parameters, and body:

```
int add(int a, int b) { return a + b; } int main() { int result = add(5, 3); std::cout << "Result: " <<
result << std::endl; return 0; }
```

Functions can have different return types:

- **void:** Returns nothing
- **int, double, etc.:** Returns a value of that type
- Functions can also accept parameters by value or by reference

Classes and Objects

C++ is an object-oriented language. Classes are blueprints for creating objects. They encapsulate data and functions that operate on that data.

```
class Rectangle { private: int width; int height; public: Rectangle(int w, int h) { width = w; height = h; }
int area() { return width * height; } }; int main() { Rectangle rect(10, 5); std::cout << "Area: " <<
rect.area() << std::endl; return 0; }
```

Key OOP concepts in C++:

- **Encapsulation:** Bundling data and methods together
- **Inheritance:** Creating new classes from existing ones
- **Polymorphism:** Objects of different types responding to the same interface
- **Abstraction:** Hiding complex implementation details

Question 4

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>The Daily Tech Tribune</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }
        body {
            font-family: 'Georgia', serif;
            background: #f5f5f5;
            padding: 30px;
        }
        .newspaper-layout {
            max-width: 1200px;
            margin: 0 auto;
            background: white;
            padding: 40px;
            box-shadow: 0 10px 40px rgba(0,0,0,0.1);
        }
        .title {
            text-align: center;
            border-bottom: 4px double #2c3e50;
            padding-bottom: 20px;
            margin-bottom: 30px;
        }
        .title h1 {
            font-size: 4em;
            font-family: 'Times New Roman', serif;
            color: #2c3e50;
            letter-spacing: 2px;
            margin-bottom: 10px;
        }
        .subtitle {
            color: #7f8c8d;
            font-size: 1.1em;
            font-style: italic;
        }
        article h2 {
            font-size: 2em;
            color: #2c3e50;
            margin-bottom: 15px;
            line-height: 1.3;
        }
        article p {
            line-height: 1.8;

            color: #34495e;
            margin-bottom: 15px;
            text-align: justify;
            font-size: 1.05em;
        }
        .feature-article {
            column-count: 2;
            column-gap: 40px;
            margin-bottom: 30px;
            padding-bottom: 30px;
            border-bottom: 2px solid #ecf0f1;
        }
        .feature-article h2 {
            column-span: all;
            font-size: 2.5em;
            color: #e74c3c;
        }
        .cover-image {
            width: 100%;
            margin: 30px 0;
        }
        .cover-image img {
            width: 100%;
            height: 400px;
            object-fit: cover;
            border-radius: 8px;
        }
```

```
        .cover-image figcaption {
            text-align: center;
            font-style: italic;
            color: #7f8c8d;
            margin-top: 10px;
            font-size: 0.95em;
        }
        .secondary-article {
            margin: 30px 0;
            padding: 30px 0;
            border-bottom: 1px solid #ecf0f1;
        }
        .secondary-article h2 {
            color: #3498db;
        }
        .articles-grid {
            display: grid;
            grid-template-columns: repeat(2, 1fr);
            gap: 30px;
            margin-top: 30px;
        }
        .small-article {
            padding: 20px;

            background: #f8f9fa;
            border-left: 4px solid #27ae60;
            border-radius: 5px;
        }
        .small-article h3 {
            font-size: 1.5em;
            color: #2c3e50;
            margin-bottom: 10px;
        }
        .small-article p {
            font-size: 1em;
        }
        .quote-box {
            background: #ecf0f1;
            border-left: 5px solid #e74c3c;
            padding: 20px;
            margin: 20px 0;
            font-style: italic;
            font-size: 1.2em;
            color: #2c3e50;
        }
        @media (max-width: 768px) {
            .feature-article {
                column-count: 1;
            }
            .articles-grid {
                grid-template-columns: 1fr;
            }
            .title h1 {
                font-size: 2.5em;
            }
        }
    </style>
</head>
<body>
    <main class="newspaper-layout">
        <!-- Header/Title -->
        <header class="title">
            <h1>■ The Daily Tech Tribune</h1>
            <p class="subtitle">Breaking News in Technology | Thursday, March 5, 2026</p>
        </header>

        <!-- Feature Article -->
        <article class="feature-article">
            <h2>AI Revolution Transforms Global Industries</h2>
            <p>Artificial Intelligence continues to reshape the landscape of modern business and technology. Compa
            <p>The impact extends beyond business, with healthcare, education, and scientific research seeing rema
            <p>"We're witnessing a fundamental shift in how humans interact with technology," says Dr. Sarah Chen
            <p>Major corporations have invested over $500 billion in AI research and development over the past yea
        </article>

        <!-- Cover Image -->
        <figure class="cover-image">
            <img src="https://images.unsplash.com/photo-1677442136019-21780ecad995?w=1200&q=80" alt="AI Technolog
            <figcaption>AI-powered systems are revolutionizing industries worldwide</figcaption>
        </figure>

        <!-- Quote Box -->
        <div class="quote-box">
            "We're witnessing a fundamental shift in how humans interact with technology. The integration of AI i
            <br><strong>— Dr. Sarah Chen, Stanford University</strong>
```

```
        </div>

        <!-- Secondary Article -->
        <article class="secondary-article">
            <h2>Quantum Computing Reaches New Milestone</h2>
            <p>Scientists at MIT have announced a breakthrough in quantum computing stability, maintaining quantu
            <p>The research team utilized a novel approach to error correction, effectively neutralizing one of q
            <p>Major tech companies including Google, IBM, and Microsoft have already expressed interest in colla
        </article>

        <!-- Small Articles Grid -->
        <div class="articles-grid">
            <article class="small-article">
                <h3>■ SpaceX Launches Record Mission</h3>
                <p>SpaceX successfully launched its 50th mission of the year, setting a new industry record. The r
                <p>CEO Elon Musk announced plans for weekly launches throughout 2026, with ambitious goals for Ma
            </article>

            <article class="small-article">
                <h3>■ Breakthrough in Cancer Treatment</h3>
                <p>Researchers have developed a new immunotherapy treatment showing 85% success rate in clinical t
                <p>FDA approval is expected by the end of the year, offering new hope to millions of patients wor
            </article>

            <article class="small-article">
                <h3>■ Climate Tech Investments Soar</h3>
                <p>Global investments in climate technology reached $150 billion in Q1 2026, marking a 300% increa
                <p>Analysts predict climate tech will become the fastest-growing sector by 2030.</p>
            </article>

            <article class="small-article">
                <h3>■ 5G Coverage Reaches 80% Globally</h3>
                <p>Telecommunications companies report that 5G network coverage now reaches 80% of the global popu
                <p>6G research has already begun, promising speeds 100x faster than current 5G networks.</p>
            </article>
        </div>
    </main>
</body>
</html>
```

Output:

The Daily Tech Tribune

Breaking News in Technology | Thursday, March 5, 2026

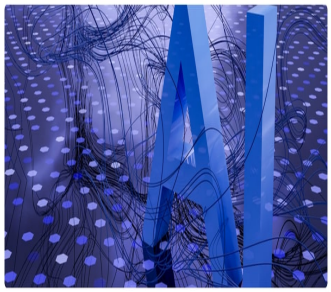
AI Revolution Transforms Global Industries

Artificial Intelligence continues to reshape the landscape of modern business and technology. Companies worldwide are reporting unprecedented efficiency gains through the implementation of advanced machine learning systems. Industry experts predict that AI-driven automation will create millions of new jobs while transforming traditional roles across all sectors.

The impact extends beyond business, with healthcare, education, and scientific research seeing remarkable breakthroughs. Machine learning algorithms are now capable of diagnosing diseases with accuracy rivaling human specialists, while educational AI tutors provide personalized learning experiences to students globally.

"We're witnessing a fundamental shift in how humans interact with technology," says Dr. Sarah Chen, leading AI researcher at Stanford University. "The integration of AI into everyday life is no longer science fiction—it's our present reality."

Major corporations have invested over $500 billion in AI research and development over the past year alone. This massive investment has yielded remarkable results, from natural language processing systems that understand context and nuance to computer vision algorithms that surpass human accuracy in medical imaging analysis.



AI-powered systems are revolutionizing industries worldwide

> *"We're witnessing a fundamental shift in how humans interact with technology. The integration of AI into everyday life is no longer science fiction—it's our present reality."*
> ***– Dr. Sarah Chen, Stanford University***

Quantum Computing Reaches New Milestone

Scientists at MIT have announced a breakthrough in quantum computing stability, maintaining quantum coherence for a record-breaking 10 minutes. This achievement brings practical quantum computers significantly closer to reality, with potential applications in cryptography, drug discovery, and climate modeling.

The research team utilized a novel approach to error correction, effectively neutralizing one of quantum computing's biggest challenges. "This is a game-changer," explains lead researcher Dr. James Morrison. "We're moving from theoretical possibility to practical implementation much faster than anticipated."

Major tech companies including Google, IBM, and Microsoft have already expressed interest in collaborating on the next phase of development. Industry analysts predict commercial quantum computers could be available within the next five years, potentially revolutionizing fields from pharmaceutical research to financial modeling.

🚀 SpaceX Launches Record Mission

SpaceX successfully launched its 50th mission of the year, setting a new industry record. The mission deployed 60 satellites for the expanding Starlink network, bringing high-speed internet to remote regions worldwide.

CEO Elon Musk announced plans for weekly launches throughout 2026, with ambitious goals for Mars exploration beginning in 2028.

💊 Breakthrough in Cancer Treatment

Researchers have developed a new immunotherapy treatment showing 85% success rate in clinical trials for previously untreatable cancers. The treatment harnesses the body's immune system to target cancer cells with unprecedented precision.

FDA approval is expected by the end of the year, offering new hope to millions of patients worldwide.

🌍 Climate Tech Investments Soar

Global investments in climate technology reached $150 billion in Q1 2026, marking a 300% increase from last year. Renewable energy, carbon capture, and sustainable agriculture lead the funding surge.

Analysts predict climate tech will become the fastest-growing sector by 2030.

📱 5G Coverage Reaches 80% Globally

Telecommunications companies report that 5G network coverage now reaches 80% of the global population. The milestone marks a new era in connectivity, enabling IoT, autonomous vehicles, and smart cities at scale.

6G research has already begun, promising speeds 100x faster than current 5G networks.

Question 5

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>TechWatch Pro - Your Smart Companion</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }
        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            background: #fff;
        }
        #header {
            background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
            color: white;
            padding: 20px 50px;
            position: sticky;
            top: 0;
            z-index: 1000;
            box-shadow: 0 2px 10px rgba(0,0,0,0.1);
        }
        .header-container {
            max-width: 1200px;
            margin: 0 auto;
            display: flex;
            justify-content: space-between;
            align-items: center;
        }
        #header-img {
            height: 50px;
        }
        #nav-bar {
            display: flex;
            gap: 30px;
        }
        .nav-link {
            color: white;
            text-decoration: none;
            font-size: 1.1em;
            transition: all 0.3s ease;
            padding: 8px 16px;
            border-radius: 5px;
        }
        .nav-link:hover {
            background: rgba(255,255,255,0.2);
        }

        #hero {
            background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
            color: white;
            text-align: center;
            padding: 100px 20px;
        }
        #hero h1 {
            font-size: 4em;
            margin-bottom: 20px;
            text-shadow: 2px 2px 4px rgba(0,0,0,0.2);
        }
        #hero p {
            font-size: 1.5em;
            margin-bottom: 40px;
            opacity: 0.95;
        }
        #form {
            display: flex;
            gap: 15px;
            justify-content: center;
            max-width: 600px;
            margin: 0 auto;
        }
        #email {
            flex: 1;
            padding: 15px 20px;
            border: none;
```

```
    border-radius: 8px;
    font-size: 1.1em;
}
#submit {
    padding: 15px 40px;
    background: #27ae60;
    color: white;
    border: none;
    border-radius: 8px;
    font-size: 1.1em;
    font-weight: bold;
    cursor: pointer;
    transition: all 0.3s ease;
}
#submit:hover {
    background: #229954;
    transform: translateY(-2px);
    box-shadow: 0 5px 15px rgba(39, 174, 96, 0.4);
}
#features, #how-it-works, #pricing {
    max-width: 1200px;
    margin: 80px auto;
    padding: 0 20px;

}
section h2 {
    text-align: center;
    font-size: 3em;
    color: #2c3e50;
    margin-bottom: 50px;
}
.features-grid {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
    gap: 40px;
}
.feature-item {
    text-align: center;
    padding: 30px;
    border-radius: 15px;
    background: #f8f9fa;
    transition: transform 0.3s ease, box-shadow 0.3s ease;
}
.feature-item:hover {
    transform: translateY(-10px);
    box-shadow: 0 10px 30px rgba(0,0,0,0.1);
}
.feature-icon {
    font-size: 4em;
    margin-bottom: 20px;
}
.feature-item h3 {
    font-size: 1.8em;
    color: #2c3e50;
    margin-bottom: 15px;
}
.feature-item p {
    color: #7f8c8d;
    line-height: 1.6;
    font-size: 1.1em;
}
#video {
    width: 100%;
    max-width: 800px;
    height: 450px;
    display: block;
    margin: 0 auto;
    border-radius: 15px;
    box-shadow: 0 10px 40px rgba(0,0,0,0.2);
}
.pricing-grid {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
    gap: 30px;

    margin-top: 50px;
}
.pricing-card {
    background: white;
    border-radius: 20px;
    padding: 40px;
    text-align: center;
    box-shadow: 0 10px 40px rgba(0,0,0,0.1);
    transition: transform 0.3s ease;
    border: 2px solid #ecf0f1;
```

```
        }
        .pricing-card:hover {
            transform: translateY(-10px);
            border-color: #667eea;
        }
        .pricing-card.featured {
            border-color: #667eea;
            background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
            color: white;
            transform: scale(1.05);
        }
        .pricing-card h3 {
            font-size: 2em;
            margin-bottom: 20px;
        }
        .price {
            font-size: 3em;
            font-weight: bold;
            margin: 20px 0;
        }
        .pricing-card ul {
            list-style: none;
            margin: 30px 0;
        }
        .pricing-card li {
            padding: 10px 0;
            border-bottom: 1px solid rgba(0,0,0,0.1);
        }
        .pricing-card.featured li {
            border-bottom-color: rgba(255,255,255,0.2);
        }
        .pricing-btn {
            padding: 15px 40px;
            background: #667eea;
            color: white;
            border: none;
            border-radius: 8px;
            font-size: 1.1em;
            cursor: pointer;
            transition: all 0.3s ease;

        }
        .pricing-btn:hover {
            background: #764ba2;
            transform: translateY(-2px);
        }
        .pricing-card.featured .pricing-btn {
            background: white;
            color: #667eea;
        }
        @media (max-width: 768px) {
            #hero h1 {
                font-size: 2.5em;
            }
            #form {
                flex-direction: column;
            }
        }
    </style>
</head>
<body>
    <!-- Header with Navigation -->
    <header id="header">
        <div class="header-container">
            <img id="header-img" src="https://via.placeholder.com/50/ffffff/667eea?text=■" alt="TechWatch Pro Log
            <nav id="nav-bar">
                <a href="#features" class="nav-link">Features</a>
                <a href="#how-it-works" class="nav-link">How It Works</a>
                <a href="#pricing" class="nav-link">Pricing</a>
            </nav>
        </div>
    </header>

    <!-- Hero Section -->
    <section id="hero">
        <h1>■ Elevate Your Lifestyle</h1>
        <p>Experience the future of wearable technology with TechWatch Pro</p>
        <form id="form" action="#" method="POST">
            <input type="email" id="email" name="email" placeholder="Enter your email address" required>
            <input type="submit" id="submit" value="Get Early Access">
        </form>
    </section>

    <!-- Features Section -->
    <section id="features">
```

```
    <h2>Premium Features</h2>
    <div class="features-grid">
        <div class="feature-item">
            <div class="feature-icon">❤■</div>
            <h3>Health Monitoring</h3>
            <p>Track your heart rate, blood oxygen, sleep quality, and stress levels with medical-grade senso

        </div>
        <div class="feature-item">
            <div class="feature-icon">■</div>
            <h3>Fitness Tracking</h3>
            <p>Monitor 100+ workout types with automatic detection. GPS tracking, pace analysis, and real-time
        </div>
        <div class="feature-item">
            <div class="feature-icon">■</div>
            <h3>Water Resistant</h3>
            <p>50-meter water resistance means you can swim, surf, or shower with confidence. Built to withsta
        </div>
        <div class="feature-item">
            <div class="feature-icon">■</div>
            <h3>Long Battery Life</h3>
            <p>Up to 7 days on a single charge with typical use. Fast charging gives you a full day's power in
        </div>
        <div class="feature-item">
            <div class="feature-icon">■</div>
            <h3>Smart Notifications</h3>
            <p>Stay connected with calls, texts, and app alerts on your wrist. Voice assistant integration fo
        </div>
        <div class="feature-item">
            <div class="feature-icon">■</div>
            <h3>Customizable Design</h3>
            <p>Choose from hundreds of watch faces and interchangeable bands. Express your style with premium
        </div>
    </div>
</section>

<!-- How It Works Section -->
<section id="how-it-works">
    <h2>See It In Action</h2>
    <iframe
        id="video"
        src="https://www.youtube.com/embed/dQw4w9WgXcQ"
        frameborder="0"
        allow="accelerometer; autoplay; clipboard-write; encrypted-media; gyroscope; picture-in-picture"
        allowfullscreen>
    </iframe>
</section>

<!-- Pricing Section -->
<section id="pricing">
    <h2>Choose Your Plan</h2>
    <div class="pricing-grid">
        <div class="pricing-card">
            <h3>Starter</h3>
            <div class="price">$299</div>
            <ul>
                <li>✓ Basic Health Monitoring</li>

                <li>✓ 50+ Workout Modes</li>
                <li>✓ 5 Days Battery</li>
                <li>✓ Water Resistant</li>
                <li>✓ 1 Year Warranty</li>
            </ul>
            <button class="pricing-btn">Buy Now</button>
        </div>

        <div class="pricing-card featured">
            <h3>■ Pro</h3>
            <div class="price">$449</div>
            <ul>
                <li>✓ Advanced Health Monitoring</li>
                <li>✓ 100+ Workout Modes</li>
                <li>✓ 7 Days Battery</li>
                <li>✓ GPS + Cellular</li>
                <li>✓ Premium Bands Included</li>
                <li>✓ 2 Years Warranty</li>
            </ul>
            <button class="pricing-btn">Buy Now</button>
        </div>

        <div class="pricing-card">
            <h3>Ultra</h3>
            <div class="price">$599</div>
            <ul>
                <li>✓ Medical-Grade Sensors</li>
```

```
                    <li>✓ Unlimited Workouts</li>
                    <li>✓ 10 Days Battery</li>
                    <li>✓ Titanium Case</li>
                    <li>✓ Sapphire Crystal</li>
                    <li>✓ Lifetime Warranty</li>
                </ul>
                <button class="pricing-btn">Buy Now</button>
            </div>
        </div>
    </section>
</body>
</html>
```

Output:

TechWatch Pro Logo

Features How It Works Pricing

⌚ Elevate Your Lifestyle

Experience the future of wearable technology with TechWatch Pro

Enter your email address Get Early Access

Premium Features

❤️

Health Monitoring

Track your heart rate, blood oxygen, sleep quality, and stress levels with medical-grade sensors. Get personalized health insights powered by AI.

🏃

Fitness Tracking

Monitor 100+ workout types with automatic detection, GPS tracking, pace analysis, and real-time coaching to help you reach your fitness goals.

💧

Water Resistant

50-meter water resistance means you can swim, surf, or shower with confidence. Built to withstand any adventure.

🔋

Long Battery Life

Up to 7 days on a single charge with typical use. Fast charging gives you a full day's power in just 30 minutes.

📱

Smart Notifications

Stay connected with calls, texts, and app alerts on your wrist. Voice assistant integration for hands-free control.

🎨

Customizable Design

Choose from hundreds of watch faces and interchangeable bands. Express your style with premium materials and finishes.

See It In Action

Choose Your Plan

Starter

$299

- ✓ Basic Health Monitoring
- ✓ 50+ Workout Modes
- ✓ 5 Days Battery
- ✓ Water Resistant
- ✓ 1 Year Warranty

Buy Now

⭐ Pro

$449

- ✓ Advanced Health Monitoring
- ✓ 100+ Workout Modes
- ✓ 7 Days Battery
- ✓ GPS + Cellular
- ✓ Premium Bands Included
- ✓ 2 Years Warranty

Buy Now

Ultra

$599

- ✓ Medical-Grade Sensors
- ✓ Unlimited Workouts
- ✓ 10 Days Battery
- ✓ Titanium Case
- ✓ Sapphire Crystal
- ✓ Lifetime Warranty

Buy Now